

Polynomial AI

Backend Engineering & GenAI Foundations

Confidential Candidate Assessment Pack | PAI/AR1/001

Overview & Guidelines

Company Overview: Polynomial AI builds enterprise AI systems, retrieval platforms, agentic workflows, and production-grade applications. This assessment evaluates software engineering, architecture design, problem-solving, scalability, security, and communication skills.

Assessment Rules: Expected duration: 48-72 hours. Candidates may use AI tools but must explain design decisions. Production quality code, documentation, and architecture are mandatory.

Technical Expectations:

Python (Any version)

FastAPI

PostgreSQL

Docker

Streamlit/Gradio

Git

Testing Frameworks

Logging

Configuration Management

Project Requirements

Candidates must complete both of the following core projects. Ensure both are production-ready and fully documented.

Project 1: GenAI Document Intelligence Platform

Core Objective: Build a document ingestion and Q&A platform supporting PDF, DOCX and TXT uploads.

Key Features: Implement chunking, embeddings, vector search, citations, conversational memory, document management and containerized deployment. Required APIs include upload, query, delete, list documents and health endpoints.

Functional Requirements: Authentication, API design, validation, persistence, error handling, logging, monitoring, deployment and documentation.

Project 2: Banking Core Operations Platform

Core Objective: Design a PostgreSQL-backed banking platform supporting KYC onboarding, account creation, account maintenance, and debit/credit transactions.

Key Features: Implement transaction history, reporting, ledger integrity, audit trails and concurrency-safe transaction processing.

Functional Requirements: Role-based access, reporting, administration, auditability, analytics, security controls and production readiness.

Architecture & Non-Functional Requirements

Mandatory Deliverables

- High-Level Architecture Diagram
- Deployment Diagram
- Component Diagram
- Database ER Diagram
- Sequence Diagrams
- API Documentation & README

Non-Functional Needs

- Scalability & Security
- Observability & Monitoring
- Resilience & Fault Tolerance
- Maintainability & Auditability
- Performance Optimization

Submission & Evaluation

Expected Artifacts: Git Repository, README, Demo Video (5-10 mins), Deployment URL (optional), Flowcharts, Sequence Diagrams, Architecture Document, and Trade-off Analysis.

Criteria	Marks	Criteria	Marks
Architecture	20	Security	10
Code Quality	15	Scalability	10
Database Design	15	Documentation	10
AI/LLM Integration	15	UI / UX	5

Bonus Challenges: CI/CD, caching, automated testing, monitoring dashboards, advanced analytics, agent workflows, performance optimization, and cloud deployment.